



A Bayesian change point analysis re-examines the 4.2 ka BP event in southeast Europe and southwest Asia

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ABSTRACT

The 4.2 ka BP event is a hypothesized global drought and cooling event and is claimed to be supported by proxy records from a wide range of geographical locations. However, recent studies have presented contradictory data that call into question the validity of this hypothetical event in certain regions, leading to renewed interest in defining its true geographical and temporal extent. This study applies a Bayesian model designed to test the existence of abrupt change points to paleoclimate time series from southeast Europe and southwest Asia, which are claimed to support this hypothesis. The results are discussed in relation to previous studies on the geographical extent of the 4.2 ka BP event. The results suggest that the evidence for the 4.2 ka BP event may not be as strong as previously thought. While this analysis supports the event that occurred in the central Mediterranean and Greece, it does not appear to have had an impact on proxy data from Anatolia and the Levant. Further proxy data are needed to confirm the existence of this potential climate event and better constrain its geographical extent.

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1. Introduction

The current human-induced climate change debate/crisis has generated significant scholarly interest in the possibility of past abrupt climatic events in the Holocene and their impact on ancient societies. The 8.2 and 4.2 ka BP¹ events, in particular, are now widely recognized and used as stratigraphic boundaries of the Middle Holocene (Walker et al., 2019). Of these, the 4.2 ka BP event has been posited as a global drought and cooling event (Weiss, 2015, 2016), based on proxy records from a wide range of geographical locations. Walker et al. (2019, p. 6) claim, "The event appears to be global in nature, occurring in proxy records across seven continents from North America and northern Europe, through the Mediterranean, Middle East and India to China and Australia; and across Africa, Andean–Patagonian South America, and Antarctica." Despite the widespread acceptance of the 4.2 ka BP event as a global phenomenon, a growing body of literature has called its validity into question in certain regions (Ön et al., 2021;

Bradley and Bakke, 2019; Kathayat et al., 2018; Jones et al., 2016; Arz et al., 2015; Roberts, 2016; Roland et al., 2014). This has led to a reevaluation of the global character of the 4.2 ka BP event and a renewed interest in understanding its true geographical and temporal extent.

For example, a stalagmite sample from Mawmluh Cave in Meghalaya, India, was selected as the stratotype of the Middle-Late Holocene boundary (Walker et al., 2019; Berkelhammer et al., 2013). However, a recent study has called into question the validity of this sample, presenting contradictory data for the timing and magnitude of the event from the same cave (Kathayat et al., 2018). Moreover, Scroxton et al. (2022) has investigated the common variability of ten different records from around the tropical Indian Ocean basin, questioning the original hypothesis and proposing seasonal climatic changes for different periods in the region. Other studies claim to show no drought event in certain regions such as Northern Ireland (Roland et al., 2014), Sofular Cave on the Black Sea coasts of Anatolia (Göktürk et al., 2011; Jones et al., 2016), Lake Hazar, on headwaters of the Tigris (Ön et al., 2017a), and the Zagros Mountains (Andrews et al., 2020). A recent temperature reconstruction of the eastern Mediterranean through a Bayesian multilevel regression model suggests, in contrast to claims of global cooling, that the region was slightly warmer than the average for

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¹ I use "ka BP" (thousand years before present) convention throughout the study, where present corresponds to 1950 CE as defined by Flint and Deevey (1962).

most of the Holocene during the temporal interval of interest (Ön et al., 2023).

These findings have led to critical reassessment of the evidence for, and geographical distribution of the 4.2 ka BP event. For example, Di Rita and Magri (2019) suggested that while there was no change in vegetation cover in northern Italy during this period, there was a noticeable decrease in forest cover in the southern Italian Peninsula and this geographical transition appears to have been gradual. Railsback et al. (2018, 2022) argued that the 4.2 ka BP event was an unusual wet period in southern Africa, contradicting claims to this being a global drought. Additionally, Bradley and Bakke (2019) assert that there is no compelling evidence for the 4.2 ka BP event in the northern North Atlantic, which is a key region for the formation of the global thermohaline circulation. A recent critical article used a hidden Markov model to analyze the time series of proxy data from southeast Europe and southwest Asia and suggested that not all data provide credible evidence for the event (Ön et al., 2021). This approach utilized a control set of time series data that did not show the event in the region, and the datasets claiming to show the event were reconstructed through the control set in a Bayesian manner. The differences between the reconstructed and original series were examined to assess the impact of the event. Unlike the present study, the hidden Markov approach used in (Ön et al., 2021) required a control set of data that showed no evidence of the hypothesized climatic event, and each set of data needed to be evenly spaced and synchronous in time.

This study used Bayesian change point analysis (Carlin et al., 1992; Gallagher et al., 2011) to evaluate the evidence for the 4.2 ka BP event in paleoclimate proxy data from southeast Europe and southwest Asia (Fig. 1). These regions are known for the emergence of complex societies and which are claimed to have been affected by the alleged abrupt climate change (Höflmayer, 2017; Dalfes et al.,

1997). The objective of this study is to detect possible abrupt changes in the time series of proxy data using a piecewise linear regression model with identified change points. The aim is to examine the implications of these findings and their relevance to previous assertions regarding the 4.2 ka BP event. The applied statistical model conducts a fully Bayesian inference process to estimate the locations of the change points for each dataset independently, as well as the parameters of the piecewise regressions between the change points, along with the rest of the model parameters. The Bayesian approach is chosen for its ability to incorporate prior beliefs and reflect uncertainty, which are superior to frequentist methods.

2. Materials and methods

2.1. Data

To apply the Bayesian change point analysis as outlined in Section 2.2, I followed selection criteria for the data inclusion similar to those used in Ön et al. (2021). The criteria were as follows:

1. The time series plot of the proxy data must demonstrate a change over the temporal period of interest, as identified by the authors of the data or any subsequent study.
2. The data must span the time period between 7.5 and 3.5 ka BP, in order to exclude other hypothetical abrupt changes such as the 8.2 ka BP event (e.g., Parker and Harrison, 2022) and the 3.2 ka BP event (e.g., Kaniewski et al., 2019).
3. In order to accurately detect an anomaly as short as 300 years, I only selected data with a temporal resolution of at least 50 years.

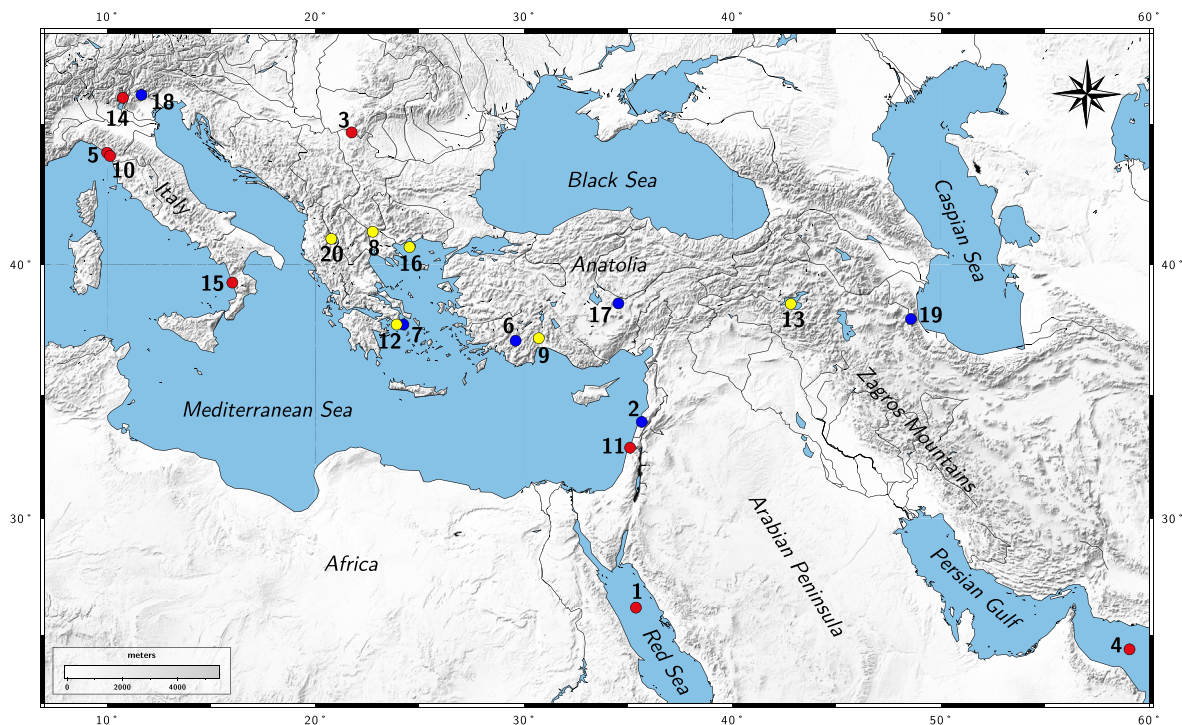


Fig. 1. Location map of the proxy data given in Table 1. The map was generated using the Generic Mapping Tools software (Wessel et al., 2019). Note that the locations of datasets #7 and #12 represent different measurements conducted by different authors using the same sediment core (Table 1), and are shown as distinct points on the map by a slight east-west displacement. The color of each location point corresponds to the colors of the results, which are explicitly explained and given in Section 3. Blue points indicate locations that show no change in the time series during the temporal interval of interest, while yellow and red points indicate locations that show one and two change points, respectively.

Table 1

The paleoclimate proxy data used in this study. Site type and other used abbreviations are as follows: M = Marine, AS=Archaeological site, S=Speleothem, L = Lake.

Location (Sample name)	Site type	Proxy	Proxy interpretation	Reference
1. Red Sea (GeoB 5836–2)	M	$\delta^{18}\text{O}$	SST	Arz et al. (2006)
2. Jeita Cave (J-1)	S	$\delta^{18}\text{O}$	Precipitation	Cheng et al. (2015)
3. Poleva Cave (PP98-10)	S	$\delta^{18}\text{O}$	Temperature	Constantin et al. (2007)
4. Gulf of Oman (M5-422)	M	CaCO_3	Aeolian depos.	Cullen et al. (2000)
5. Buca della Renella (RL4)	S	$\delta^{18}\text{O}$	Precipitation	Drysdale et al. (2006)
6. Gölhisar (GHA)	L	$\delta^{18}\text{O}$	Water balance	Eastwood et al. (2007)
7. Aegean Sea (GeoTü SL148)	M	Smect./Ill.	Drought	Ehrmann et al. (2007)
8. Lake Dojran (Co1260)	L	CaCO_3	Productivity	Francke et al. (2013)
9. Kocain Cave (Ko-1)	S	$\delta^{13}\text{C}$	Winter temp.	Göktürk (2011)
10. Corchia Cave (CC27)	S	$\delta^{18}\text{O}$	Precipitation	Isola et al. (2019)
11. Tel Akko (Akko core)	AS	pollen	Precipitation	Kaniewski et al. (2013)
12. Aegean Sea (GeoTü SL148)	M	$\delta^{13}\text{C}_{\text{Um}}$	Productivity	Kuhnt et al. (2008)
13. Lake Van (Van 90–10)	L	quartz	Aeolian depos.	Lemcke and Sturm (1997)
14. Lake Ledro (LL081)	L	pollen	Summer precip.	Peyron et al. (2013)
15. Lake Trifoglietti (S2)	L	pollen	Summer precip.	Peyron et al. (2013)
16. Skala Marion Cave (MAR_L)	S	$\delta^{18}\text{O}$	Precipitation	Psomiadis et al. (2018)
17. Eski Acıgöl (ESK96-97)	L	$\delta^{18}\text{O}$	Water balance	Roberts et al. (2001)
18. Grotta di Ernesto (ER76)	S	$\delta^{13}\text{C}$	Temperature	Scholz et al. (2012)
19. Neor Lake	L	XRF-Ti	Aeolian depos.	Sharifi et al. (2015)
20. Lake Ohrid (Lz1120)	L	CaCO_3	Productivity	Wagner et al. (2009)

By applying these criteria, the number of eligible datasets has been restricted, particularly within the relatively small geographical region used in this study. Therefore, the data used as response variables by Ön et al. (2021) were also utilized in this study (Table 1). To the best of my knowledge, the Corchia Cave study by Isola et al. (2019) is the only study that meets the criteria outlined in the previous paragraph and was not included in Ön et al. (2021). This dataset was published while our previous article was under review and is now included in the current study.

Using data with high resolution and relatively high noise levels can pose challenges in achieving convergence of Gibbs sampling. The difficulty in achieving convergence for high-resolution data may be due to the complexity of the model and the large number of data points. Additionally, the posterior distributions of change points of high-resolution data may possibly be multi-modal, making it difficult for Gibbs sampling to converge. To address this issue, time series data with observation lengths exceeding 200 were resampled to a mean resolution of around 30 years (Table S1).

Resampling reduces the complexity of the model, the number of data points, and hence, reduces a potential multi-modality of the posterior distribution, enhancing convergence. Furthermore, the resampling procedure decreases the computational load, which lowers the risk of numerical instabilities and improves the ease of convergence. Despite the potential loss of information from resampling, a visual comparison of the resampled and original datasets indicates that the general trends and potential jumps in the time series are preserved (Fig. 2).

A first-order Savitzky-Golay filter (Savitzky and Golay, 1964) was applied to smooth the time series. This method has been used in previous paleoclimate studies (Ön et al., 2017a, 2017b, and Özeren, 2018), and it is an appropriate method when using data that are unevenly spaced in time. The sliding window lengths used for the different datasets were 5 for Jeita Cave $\delta^{18}\text{O}$ (Cheng et al., 2015), 9 for Kocain Cave $\delta^{13}\text{C}$ (Göktürk, 2011), 3 for Corchia Cave $\delta^{18}\text{O}$ (Isola et al., 2019), 7 for Grotta di Ernesto $\delta^{13}\text{C}$ (Scholz et al., 2012), and 9 for Neor Lake XRF-Ti counts (Sharifi et al., 2015). The resampling process began with the midpoint of the first window, with subsequent resampling every 5th, 9th, 3rd, 7th and 9th data point for Jeita Cave, Kocain Cave, Corchia Cave, Grotta di Ernesto and Neor Lake data, respectively.

2.2. Bayesian change point analysis

Given a standard univariate proxy time series represented by (x_i, y_i) , where y_i is the proxy data and x_i is the corresponding time points defined by an age model that are not necessarily evenly spaced, an assumption of m change points can be made to partition the time series into $m + 1$ segments. The locations of the change points, which can be objectively estimated within the model between the initial and terminal time points of the time series, allow for an assumption of a piecewise regression model $\hat{y}_i = \alpha_k + \beta_k(x_i - c_k)$ for each (x_i, y_i) pair in each segment, where $\hat{y}$ denotes the predicted value of y . Here, c_k is the location of the k th change point, and this model forms the basis of an assumption of piecewise linear regressions before and after each change point.

In each segment and for each standardized (x_i, y_i) , the model assumptions for the likelihood in this study are as follows:

$$y_i \sim t_\nu(\mu_i, \sigma^2) \quad (1)$$

where,

$$\mu_i = \alpha_{[J_i^1 + \dots + J_i^m]} + \beta_{[J_i^1 + \dots + J_i^m]}(x_i - c_{[J_i^1 + \dots + J_i^m]}), \quad (2)$$

and

$$J_i^1 = I(x_i \leq c_1), \quad (3)$$

$$J_i^2 = I(c_1 < x_i \leq c_2), \quad (4)$$

$$\vdots$$

$$J_i^m = 1 + I(x_i > c_k), \quad (5)$$

where $I(x)$ is the indicator function that equals 1 if x is true, and 0 otherwise. The t -distribution used is denoted by t_ν , where ν represents the degrees of freedom (or normality parameter). The t -distribution is parametrized by the normality parameter ν , as well as the location parameter μ and the scale parameter σ , as also described in some fundamental Bayesian statistics references such

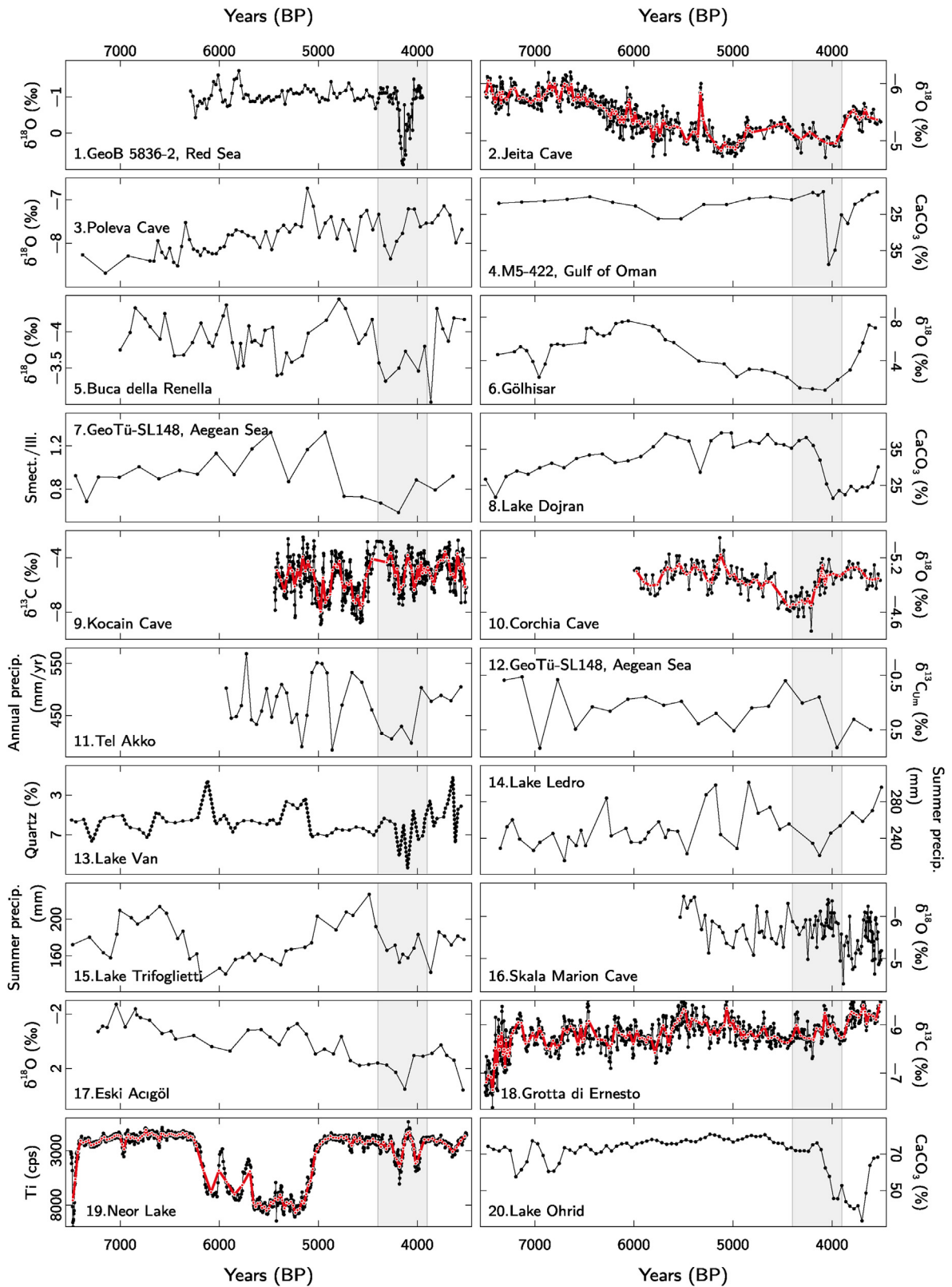


Fig. 2. Time series plots of the data used in the test. The gray shaded area indicates the (4.4, 3.9) ka BP temporal interval. The red curves over some of the time series represent filtered-sampled data as explained in Section 2.1.

as Gelman et al. (2013) and Kruschke (2015), and as implemented in the JAGS Gibbs sampler program (Plummer, 2003). The degrees of freedom parameter ν is named as normality parameter, because as it increases, the shape of the t -distribution approaches to the normal distribution. Note that when there is only one assumed change point, the indicator variable J_i^1 reduces to $1 + I(t_i > c_1)$. The same priors are used for all α and β values and are defined as follows:

$$\alpha \sim N(0, 10), \quad (6)$$

$$\beta \sim N(0, 1), \quad (7)$$

$$\sigma \sim U(10^{-3}, 10^3), \quad (8)$$

$$\nu \sim \text{Exp}(1/30). \quad (9)$$

The priors for the change points, c_k in Equation (2), vary according to the number of presumed change points. The prior distributions listed below were selected based on the number of change points, respectively.

$$c_1 \sim N^{(x_1, x_n)}(4400, 400^2), \quad (10)$$

$$c_2 \sim N^{(x_1, x_n)}(3800, 400^2), \quad (11)$$

$$c_3 \sim U(x_1, x_n), \quad (12)$$

$$c_4 \sim U(x_1, x_n), \quad (13)$$

where x_1 and x_n are the initial and terminal time points, respectively. Note that in the above equations, N denotes normal distribution, $N^{(x_1, x_n)}$ denotes truncated normal distribution with truncation range (x_1, x_n) . The mean and standard deviation values for the normal distributions in Equations (10) and (11) were first standardized with respect to each age model of the analyzed data prior to their incorporation into the model. U denotes uniform distribution, and Exp denotes exponential distribution.

The linear regression models in each segment are described using t -distribution (Equation (1)), which is robust to outliers. The location parameter μ_i is defined as $\alpha_k + \beta_k(x_i - c_k)$ (Equation (2)). The scale parameter is defined by a uniform distribution (Equation (8)). The parameter for normality is specified as an exponential distribution with a mean value of 30, as shown in Equation (9). Setting the mean of ν as 30 results in an equal allocation of weight to values above and below 30, thereby assigning equal probability to a heavy-tailed t -distribution and a nearly normal distribution. α terms, are assumed to obey a vague prior, as given in Equation (6). The slope parameters, β 's are described using a moderate prior (Equation (7)). The reason for providing a relatively moderate prior is to avoid possible steep slopes for temporally scarce data. For the temporal locations of the change points, if there is hypothetically only a single change point, the model uses the vague prior given in Equation (10). If there are two change points, the model uses the priors given in Equations (10) and (11). If there are assumed to be more change points, uninformative priors are used for the temporal locations of the rest of the change points given in Equations (12) and (13).

In this Bayesian change point model, the value of a change point, denoted as c_k , is a parameter to be estimated from the data. The model defines a prior distribution for the value of c_k , and uses the observed data to update this prior distribution to a posterior

distribution. The likelihood function specifies the probability of observing the data given a particular value of c_k and other model parameters. In this case, the likelihood function is defined as a t -distribution with location parameter μ , scale parameter σ , and normality parameter ν . The model fits the likelihood function to the observed data for each observation i , by calculating the probability of observing y_i given μ_i , σ , and ν . The value of μ_i is determined by the value of c_k , and a sequence of binary indicators, J_i^k . The binary indicators determine which linear regression equations to use for each data point. To estimate the value of a c_k that maximizes the likelihood of the observed data, the Markov chain Monte Carlo (MCMC) algorithm explores the posterior distribution of c_k , which is proportional to the product of the prior distribution and the likelihood function. The algorithm generates a sequence of samples from the posterior distribution, and uses these samples to estimate the posterior distribution and the uncertainty interval of c_k . The value of c_k that maximizes the likelihood is selected as the median of the posterior distribution. The JAGS model ensures that in a multiple change point model configuration, the change points are ordered by sorting the estimated change points in ascending order after defining the change point priors and before updating the likelihood at each step of the loop. The point estimates are obtained using the median values of the posterior distributions, while the 89% highest density intervals (HDIs) are used to quantify the uncertainty of the results.

To select the most appropriate change point model, I employed the use of widely applicable information criteria (WAIC) as a measure of predictive accuracy, following the approach proposed by Watanabe (2010). This method is based on the posterior mean of the likelihood and posterior variance of the log-likelihood, and is computed as follows:

$$\text{WAIC} = -2 \sum_{i=1}^n \log f_i + 2p_W \quad (14)$$

where f represents the posterior mean of the likelihood for the i th observation, y_i , and p_W is the sum of the variances of the log-likelihoods for each observation (Reich and Ghosh, 2019). While WAIC is not perfect, it is a fast, convenient, and a fully Bayesian

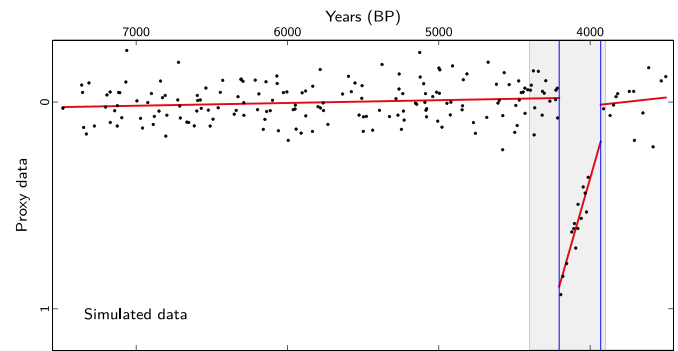


Fig. 3. The figure illustrates an example of the change point problem that is primarily considered in this study. The simulated data spans the period between 7.5 and 3.5 ka BP and data points are unevenly spaced in time. Prior to 4.2 ka BP, and after 3.9 ka BP, the data is modeled as Gaussian noise with a mean of 0 and a standard deviation of 0.1. During the interval between 4.2 and 3.9 ka BP, the data is represented as an inclined linear regression, displaying an impulse at 4.2 ka BP, such that the value of the simulated data increases to 1 and subsequently linearly attenuates to 0 until 3.9 ka BP. The noise is modeled as Gaussian with a standard deviation of 0.1. The priors used in this analysis are identical to those described in Section 2.2. The gray shaded area indicates the (4.4, 3.9) ka BP temporal interval, and the vertical blue lines depict the change points identified by the change point algorithm using a two change point assumption. It is noteworthy how the algorithm accurately detects the change points.

approach that makes it a better alternative to other model comparison methods, as noted by Gelman et al. (2014).

Gibbs sampling by the program JAGS (Plummer, 2003) through the interface *runjags* (Denwood, 2016) in R (R Core Team, 2023) was used to find posterior distributions of the model. In JAGS, the Gibbs sampling process was carried out for the standardized data model. The first 5000 iterations were discarded as adaptation steps and another 5000 iterations were discarded as burn-in steps. Four different MCMC chains with random initial points were run for each dataset and each change point assumption, with each chain consisting of 200 000 iterations. I ensured convergence of the Bayesian model by assessing the convergence diagnostics of each model and dataset using a combination of trace plots, autocorrelation plots, Gelman-Rubin statistics, and density plots, stacked for all four runs for all parameters. If MCMC runs do not seem to converge because of valley-like structures in the parameter space (Kruschke, 2015), then the adaptation, burn-in and iteration steps are increased until convergence is established.

3. Results and discussion

The change point algorithm is a powerful tool that can be applied in various ways to accommodate different model configurations. One of the simplest methods is to use constant levels on either side of presumed change points, however, this approach disregards the trends that can exist within each segment of data defined by the change points, potentially leading to an incomplete analysis. Secondly, the algorithm can be applied to force the model to be continuous at each point, as demonstrated in the work of Cahill et al. (2015). Even though such an approach allows for continuous transitions through the change points, the resolution of paleoclimate data is often low, and the manifestation of an abrupt event, if any, is typically in the form of an impulse at the time of the change point, followed by a gradual attenuation, as exemplified in Fig. 3. Some notable examples of the 4.2 ka BP event that explicitly exhibit such impulse geometry include those reported by Arz et al. (2006) and Cullen et al. (2000). These datasets can be observed as labeled numbers 1 and 4 in Fig. 2.

In order to replicate this impulse-like geometric structure (Fig. 3) observed in some notable examples of the 4.2 ka BP event, I have chosen to implement piecewise linear regressions between each identified change point. This approach does not exclude the possibility of a constant level or a continuous smooth transition approach, however, it is most likely expected that the results will give linear trends with discontinuities at each change point. A prime example of a single impulse followed by an attenuation should include two change points, assuming that the data were to return to the “normal” state prior to the impulse. This has been simulated, and the results of the change point analysis are shown in Fig. 3. However, some data may only have a single change point where an impulse event results in an enduring effect as reported in studies such as those by Francke et al. (2013). The analyzed data may exhibit three change points, as shown in the analysis of the (1) Red Sea data in Fig. 5. Data may potentially exhibit an additional change point outside of the typical two change point geometry within the temporal interval of interest or even show four or more change points. To account for such possibilities, I applied the analysis for change points ranging from one to four and tested each dataset to determine if they exhibit at least two change points or one change point that satisfies appropriate conditions (i.e., marking a possible enduring shift) within the temporal interval of interest.

Each time series was therefore analyzed for 1, 2, 3 and 4 possible change points. As discussed in Section 2.2, not all of the MCMC runs converge, those which do not, are shown by empty cells in Table 2. In some cases, multiple numbers of change point analyses may

converge. However, some of the converged runs provide very wide posterior distributions for the location of the change points (c_k in Equation (2)), resulting in high uncertainty levels (e.g., Fig. S47) and as a consequence these intervals may even overlap (e.g., Fig. S41). Alternatively, the posterior distributions of the change point locations may be bimodal (e.g., Fig. S35), or even multi-modal (e.g., Fig. S37). In this paper, multi-modality arises from different factors. One of these factors is that the presumed number of change points may not be sufficient to describe the nature of the data (e.g., Fig. S28). Another factor contributing to multi-modality is that as the number of change points increases in the discontinuous piecewise linear regression model setup, the model becomes more complex, and the parameter space becomes more fragmented. With more change points, multiple configurations of the change points may fit the data equally well. Conversely, the modes of change points in the posterior distributions may overlap due to wide posterior distributions, indicating overlapping change points (e.g., Fig. S33). This can result in multi-modal posterior distributions, where the modes correspond to less credible configurations of a probable fragmented data and change points setup. Overall, bimodality, or even multi-modality in the posterior distributions of change point locations can indicate uncertainty in the model or in the data. Therefore, it is crucial to carefully examine the posterior distributions and consider multiple model configurations when interpreting the results.

I selected a model from the set of convergent models by comparing their WAIC values for each dataset (Table S2), where a smaller WAIC value indicates a better fit. In some cases, the model with the lowest WAIC value did not have a well-defined change point model. In these instances, I selected a model based on a comparison of the posterior distributions of the change points from different models. Specifically, I selected a model that exhibited relatively narrow, unimodal posterior distributions for the change points. Additionally, I preferred models where the uncertainty intervals of the change points did not overlap with each other. Alternative change point models, with different numbers of change points, are shown in Figs. S55–S58. These figures illustrate that changing the selected model does not have a significant impact on the results of the study, particularly with respect to the test conducted on the 4.2 ka BP event.

It is possible to incorporate uncertainties arising from the nature of the proxy data, the laboratory measurements, the dating techniques, as well as the corresponding age models of each record into the Bayesian model. However, quantifying the scale of uncertainties associated with proxy data presents a challenge (McClelland et al., 2021), and such uncertainties are often not reported. Additionally, reconstructing the uncertainties related to constructed chronologies depends on factors such as the dating technique, dated material, and environment (Walker, 2005; Akçer-Öñ et al., 2020; Magee et al., 2009) and the model used to reconstruct the chronology (Trachsel and Telford, 2017). Nonetheless, none of the studies utilized in this research reported an uncertainty for the proxy data presented, and not all the datasets employed in this study reported an uncertainty for the chronology used. Only five out of twenty studies reported chronological uncertainties utilizing completely different modeling techniques. These datasets are from Isola et al. (2019), Peyron et al. (2013) (Lake Ledro and Trifoglietti), Psomiadis et al. (2018), and Scholz et al. (2012). Additionally, it is possible to incorporate random uncertainty levels within the model. However, doing so would deviate from the original datasets that support the 4.2 ka BP climate theory, and would introduce unnecessary complexity to the problem. Hence, I made the decision to exclude any uncertainties stemming from the proxy data and chronologies, and instead conducted the analysis using the original dataset without incorporating additional uncertainty.

Table 2

The table provides a summary of the median values of the posterior distributions obtained from the change point analysis for each dataset. The intervals in parentheses indicate the 89% HDI of the posterior distribution for each change point, designated as CP-1, CP-2, CP-3, and CP-4. The upright values, provided only for one change point per location, indicate the selected values for the change points for that particular location, as depicted in Figs. 5 and 4. The cells filled with a cross (×) indicate instances where the MCMC algorithm did not converge. The cells with italicized values are runs that were not selected due to the reasons explained in the main text.

Location	CP-1	CP-2	CP-3	CP-4
1. Red Sea	4200(4205, 4185)	4190(4195, 4185) 4045(4060, 4040)	4180(4190, 4175) 4115(4120, 4110) 4045(4050, 4040)	×
2. Jeita Cave	5150(5255, 4870)	×	6590(6925, 6365) 4890(4935, 4870) 3880(3945, 3840)	×
3. Poleva Cave	4570(5330, 4230)	4370(4830, 4270) 4135(4270, 3690)	4735(6740, 4270) 4290(4740, 4030) 4095(4205, 3610)	×
4. Gulf of Oman	4330(5240, 3540)	4060(4090, 4040) 3940(3970, 3910)	5790(6100, 4030) 4060(4090, 4030) 3930(3970, 3900)	×
5. Buca della Renella	4050(4630, 3630)	4450(5130, 3900) 3830(3990, 3560)	5040(5640, 4320) 4420(4720, 3840) 3830(3860, 3670)	×
6. Gölhisar	5400(5600, 5100)	5570(5770, 5340) 3780(3890, 3720)	5900(6490, 5390) 4050(5400, 3760) 3760(3820, 3650)	×
7. Aegean Sea (Semct./Ill.)	4830(4910, 4740)	4830(4920, 4740) 4100(4730, 3640)	4900(6670, 4740) 4740(4930, 4020) 4060(4500, 3640)	×
8. Lake Dojran	4105(4170, 4055)	5010(5780, 4570) 4090(4120, 4060)	5400(5930, 4740) 4210(5300, 4060) 4070(4120, 3570)	×
9. Kocain Cave	4520(4560, 4460)	4520(4550, 4460) 3800(4060, 3520)	4620(5080, 4460) 4480(4580, 3870) 3790(4030, 3520)	×
10. Corchia Cave	4160(4210, 4130)	×	5060(5720, 4950) 4530(4610, 4210) 4155(4200, 4115)	×
11. Tel Akko	4440(4660, 4120)	4443(4665, 4113) 3975(4102, 3560)	4690(5480, 4190) 4340(4600, 3890) 3960(4060, 3560)	4650(5500, 4000) 4390(4920, 3890) 4070(4530, 3620) 3870(4120, 3560)
12. Aegean Sea ($\delta^{13}\text{C}_{\text{Um}}$)	4260(4990, 3750)	4527(5032, 3925) 4017(4295, 3610)	5330(7310, 4430) 4260(4920, 3880) 3990(4210, 3610)	6790(7310, 4320) 4500(6920, 3940) 4080(4600, 3700) 4000(4190, 3610)
13. Lake Van	4250(4330, 4200)	×	×	×
14. Lake Ledro	4320(4820, 3500)	4427(5438, 4127) 3893(4251, 3500)	5280(6650, 4130) 4330(4960, 3670) 3850(4180, 3500)	5350(5580, 4240) 4910(5400, 4260) 4380(4960, 4030) 4120(4430, 3650)
15. Lake Trifoglietti	5090(5380, 4770)	×	6330(6390, 6230) 5010(5110, 4310) 4350(4490, 3860)	6540(6760, 6300) 6150(6370, 5800) 4420(4740, 4310) 4310(4440, 3780)
16. Skala Marion Cave	4470(4770, 3920)	3930(3990, 3680) 3890(4250, 3500)	4710(5390, 4250) 3930(4020, 3670) 3690(3790, 3510)	×
17. Eski Acıgöl	4640(5120, 3760)	4640(5100, 3970) 3630(4120, 3540)	5090(6780, 4270) 4120(4770, 3610) 3610(4080, 3540)	×
18. Grotta di Ernesto	4930(5060, 4850)	5040(5610, 4910) 3860(4420, 3790)	×	×
19. Neor Lake	×	6170(6220, 6080) 5060(5090, 5030)	×	×
20. Lake Ohrid	4060(4110, 4020)	×	×	×

The original theory put forth by Weiss et al. (1993) states that the drought event began around 4.2 ka BP and ended around 3.9 ka BP, although subsequent publications have extended the duration of this interval in both directions. Recently, Zanchetta et al. (2018) constrained the interval of a potential event for the central Mediterranean to between 4.4 and 3.9 ka BP. I therefore assume that the onset of the climatic event as 4.4 ka BP and its termination as 3.9 ka BP, following Zanchetta et al. (2018) and Ön et al. (2021), and call this interval as the temporal interval of interest.

Selected change point analysis results of the (1)Red Sea (Arz et al., 2006), (2)Jeita Cave (Cheng et al., 2015), (4)Gulf of Oman (Cullen et al., 2000), (6)Gölhisar (Eastwood et al., 2007), (7)Aegean Sea-Semct./Ill. (Ehrmann et al., 2007), (13)Lake Van (Lemcke and Sturm, 1997), (19)Neor Lake (Sharifi et al., 2015), and (20)Lake Ohrid (Wagner et al., 2009) datasets show relatively narrow posterior distributions. On the contrary, the selected results from (3) Poleva Cave (Constantin et al., 2007), (5)Buca della Renella (Drysedale et al., 2006), (8)Lake Dojran (Francke et al., 2013), (9)

Kocain Cave (Göktürk, 2011), (10)Corchia Cave (Isola et al., 2019), (11)Tel Akko (Kaniewski et al., 2013), (12)Aegean Sea- $\delta^{13}\text{C}_{\text{Um}}$ (Kuhnt et al., 2008), (14)Lake Ledro (Peyron et al., 2013), (15)Lake Trifoglietti (Peyron et al., 2013), (16)Skala Marion Cave (Psomiadis et al., 2018), (17)Eski Acıgöl (Roberts et al., 2001) and (18)Grotta di Ernesto (Scholz et al., 2012) datasets show relatively wide posterior distributions (see, Table 2, Fig. 4, and Figs. S1–S20).

Based on the median values of the posterior distributions of the change points, (1)Red Sea, (3)Poleva Cave, and (4)Gulf of Oman datasets show two change points between 4.4 and 3.9 ka BP, and between these change points each dataset indicate a drought or cooling as stated by the original hypothesis. However, when the uncertainty intervals of the posterior distributions described by the HDI are considered, (10)Corchia Cave, (11)Tel Akko, (14)Lake Ledro, and (15)Lake Trifoglietti datasets also support the hypothesis by showing two change points in the interval of interest. Interestingly,

according to the HDI results of (5)Buca della Renella data, which is one of the most important examples for the proponents of the hypothesis, show only one change point in the temporal interval of interest. However, there is a second change point, very close to the end member of the temporal interval, i.e., 3.9 ka BP (Table 2, Fig. 4). This timing was already noted as 3.8 ka BP by Drysdale et al. (2006) and Zanchetta et al. (2016). Furthermore, considering the uncertainty intervals of the results of the (12)Aegean Sea- $\delta^{13}\text{C}_{\text{Um}}$ and (16)Skala Marion Cave data, although the results show two change points in the specified interval, the datasets indicate a wetter period in the interval of interest, which is contrary to one would expect based on a hypothesized drought period (Fig. 3). With the exception of the Aegean Sea- $\delta^{13}\text{C}_{\text{Um}}$ and the Skala Marion Cave datasets, I display the datasets mentioned in this paragraph as red points on the map (Fig. 1).

The results of the datasets from (8)Lake Dojran, (13)Lake Van,

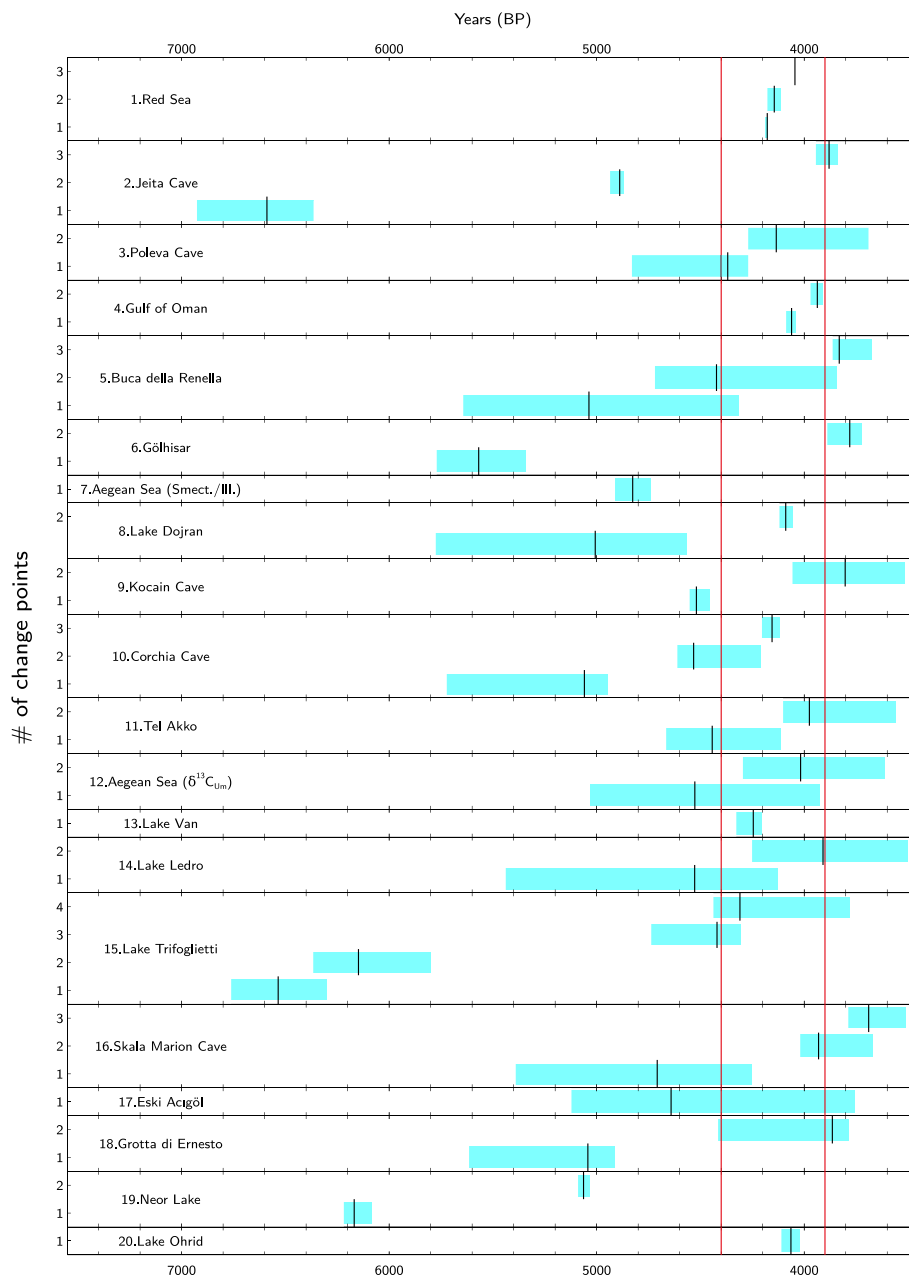


Fig. 4. The visualization of the 89% HDI of each selected change point model, given in Table 2, for the datasets given in Table 1.

and (20)Lake Ohrid show only one change point within the temporal interval of interest. The uncertainty intervals of the results from (2)Jeita Cave, (9)Kocain Cave, (17)Eski Acıgöl, and (18)Grotta di Ernesto datasets suggest that these may have a change point within the interval of interest. However, it should be noted that the results from the Jeita Cave, and Grotta di Ernesto datasets only marks a possible end point, and the uncertainty interval for the result from Eski Acıgöl data is very wide, covering the range (5120,3760) years BP. Furthermore, based on the observed environmental deterioration in their proxy data, both the Aegean Sea- $\delta^{13}\text{C}_{\text{um}}$ and Skala Marion Cave datasets may be interpreted to have a single change point at the end of the interval of interest. Except for the Jeita Cave, Eski Acıgöl, and Grotta di Ernesto data points, the datasets mentioned in this paragraph are displayed as yellow points on the map (Fig. 1).

The results of the datasets from (6)Göhlhisar, (7)Aegean Sea-Smect./Ill., and (19)Neor Lake show no change points within the temporal interval of interest. Together with the Jeita Cave, Grotta di Ernesto, and Eski Acıgöl data, they are displayed as blue points on the map (Fig. 1).

Although the change point analyses conducted in this study do not provide information regarding the uncertainties associated with the age models of each record, it is still possible to discuss the chronological uncertainties of datasets that display no or only one change point within the temporal interval of interest, as shown by the blue and yellow points in Fig. 1. Of the datasets analyzed, (2) Jeita Cave, (6)Göhlhisar, (7)Aegean Sea-Smect./Ill., (18)Grotta di Ernesto, and (19)Neor Lake datasets did not exhibit any credible change points within the temporal interval of interest. Even with chronological uncertainties considered, it is unlikely that the occurrence of a drought/cooling event proximal to the boundary points of the temporal interval could be deemed credible for these datasets. The change points of (9)Kocain Cave and (17)Eski Acıgöl emerge just before the 4.4 ka BP interval boundary. If these change points fall outside of the interval due to chronological uncertainties, they may still be considered as supporting the event. Similarly, the change points of (8)Lake Dojran, (12)Aegean Sea- $\delta^{13}\text{C}_{\text{um}}$, (16)Skala Marion Cave, and (20)Lake Ohrid are close to the 3.9 ka BP boundary of the temporal interval of interest. These datasets may be eliminated from the set of supporting datasets if their proximities are a result of chronological uncertainties.

Based on the robust chronologies of Kocain and Skala Marion Caves, which cover approximately 2000 years within the period of this study and were established using ten and nine U–Th dates, respectively, it can be concluded that they do not support either a model with two change points between 4.4 and 3.9 ka BP or a model with a single change point with a credible level shift starting at around 4.4 ka BP. In contrast, the Eski Acıgöl sediment core relies on a combination of U–Th dates for the age–depth model, with only one falling in the study period. The model assumes a constant sedimentation rate, leading to a significant chronological uncertainty. Moreover, the location of the possible change point identified from the change point analysis in this dataset has a high degree of uncertainty. As a result, the Eski Acıgöl dataset does not provide sufficient evidence to confirm the 4.2 ka BP hypothesis. Similarly, the sediment core (namely, SL148) of the Aegean Sea- $\delta^{13}\text{C}_{\text{um}}$ data has only one radiocarbon age control point in the studied interval, and the age–depth model is based on a constant sedimentation rate, raising questions about the chronology's accuracy. Additionally, chronologies of the Lake Dojran and Lake Ohrid sediment cores rely on just three radiocarbon dates within the study interval and use a polynomial interpolation approach that reports no uncertainty except for the dated depths. As a result, the chronologies of these

datasets are questionable due to the lack of uncertainty estimation. Nevertheless, despite this limitation, the presence of an impulse in the proxy record of these datasets, indicated by a change point with a narrow posterior distribution, can still be regarded as evidence supporting the 4.2 ka BP hypothesis.

Previous research on the topic of a potential climatic event at the 4.2 ka BP temporal interval has produced a range of findings. Some studies, such as reviews by Weiss (2015, 2016), did not address any spatial distributional impact of the event, instead stating that it was an “abrupt global megadrought and cooling at ca. 4.2–3.9 ka BP” (Weiss, 2016, p. 62). Other more comprehensive analyses aimed to understand and interpret the spatial and temporal distribution of the process. The review by Bini et al. (2019) collates a range of records with different resolutions and climatic interpretations across the entire Mediterranean region in order to identify a simplified pattern for the event. However, their results suggest that while a climatic event is evident, its manifestation varies geographically. The findings of this review are supported by those of Arz et al. (2015) and Sirocko (2015), where all these studies acknowledge that the available data at the time of their publications were insufficient to resolve a probable pattern.

Further studies on the 4.2 ka BP event in the central Mediterranean have suggested that the impact of this climate event may differ between the northern and southern regions (Magny et al., 2013; Di Rita and Magri, 2019). Specifically, these studies propose that the northernmost central Mediterranean did not experience significant changes during the period of interest, while southern Italy showed a decline in forest cover. Accordingly, Di Rita and Magri (2019) argue that this phenomenon may be attributed to the region's vulnerability to decreased precipitation resulting from the northward displacement of the subtropical high-pressure system over North Africa.

Di Rita et al. (2022) focused on the tree cover of the entire Mediterranean region, and claim a nonuniform spatiotemporal pattern as a reflection of the 4.2 ka BP event. They state that the western and eastern Mediterranean as a group exhibit a behavior that is quite the opposite of the central Mediterranean. Specifically, while the central Mediterranean experienced drought between 4.5 and 4.2 ka BP, the eastern and western Mediterranean experienced wetter conditions. Conversely, between 4.2 and 3.9 ka BP, the conditions were quite the opposite of the previous period.

In the study conducted by Ön et al. (2021), a Bayesian test based on a hidden Markov model was used to investigate the potential impact of a climatic event in Southeast Europe and Southwest Asia. The findings indicated that the examined proxy sites did not exhibit a clear and consistent pattern. However, the authors also proposed a probable hypothetical geographical boundary dividing the directly affected and unaffected datasets by a drought event and proposed that the datasets located south of this boundary (Fig. 6) were affected by this event. Furthermore, the study proposed that the geographic distribution of the impact might have been caused by an asymmetrical northward migration of the high-pressure system over North Africa, which was due to an asymmetric bending of the Intertropical Convergence Zone (ITCZ). As per this analysis, the asymmetric northward migration carried the hyper-arid system of the central Sahara, leading to drier conditions during the possible 4.2 ka BP drought event in the central Mediterranean and, to some extent, in the Levant. However, the available data from the Eastern Mediterranean, Anatolia, and Mesopotamia did not show direct evidence of a drought event. Instead, it is suggested that they were indirectly affected by possible intensified wind storms that were generated by the aridity in the central Mediterranean and, to some extent, the Levant.

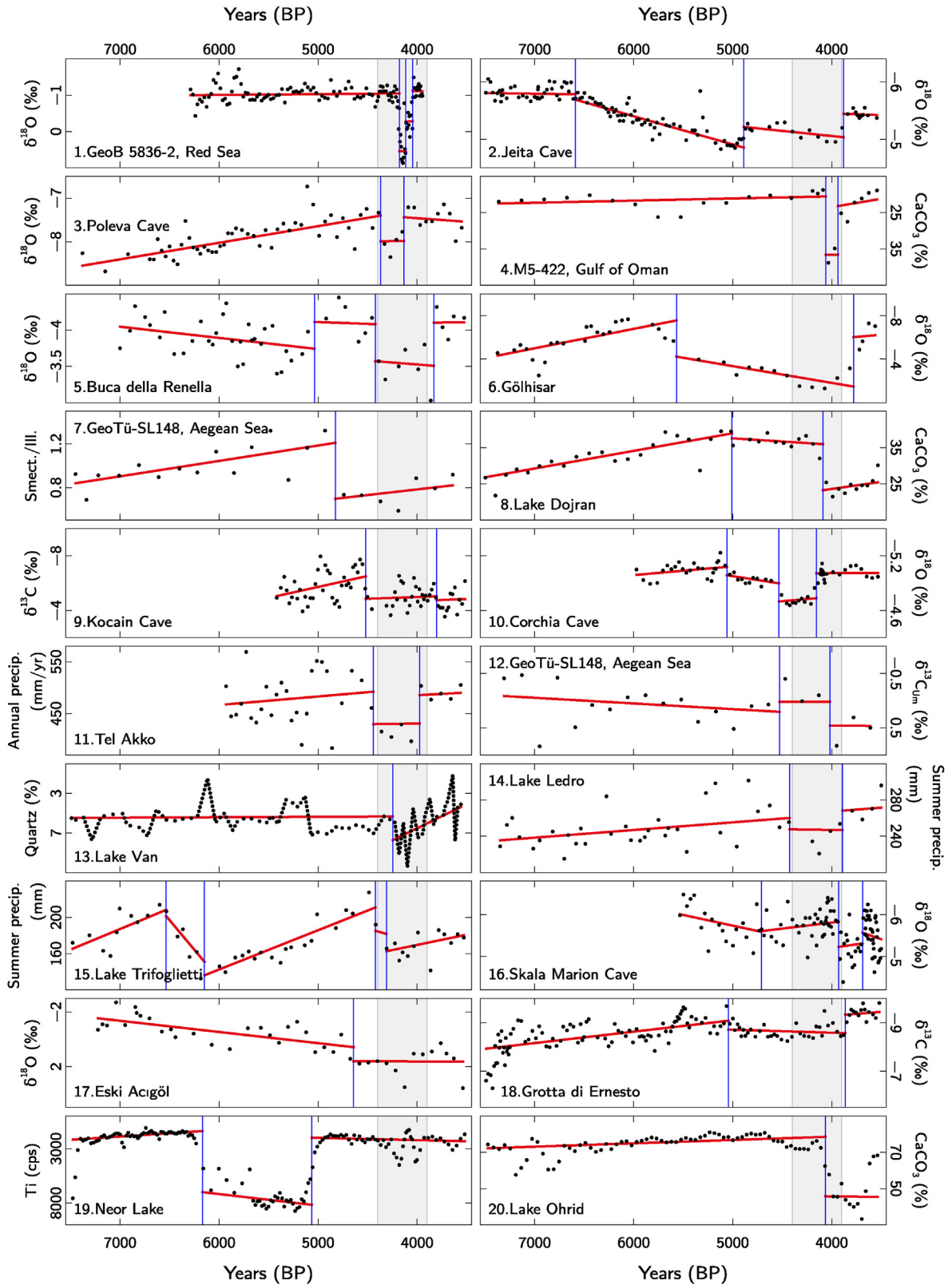


Fig. 5. Plots of the change points and the regression lines between these change points obtained by choosing the median values of the posterior distributions as representative of the results. The gray shaded area indicates the (4.4,3.9) ka BP temporal interval.

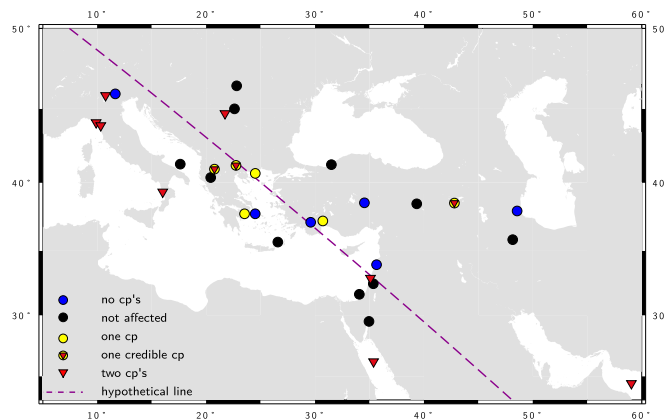


Fig. 6. The figure illustrates the spatial distribution of data points, represented by blue, yellow points and red triangles, which correspond to zero, one, and two change points between 4.4 and 3.9 ka BP, respectively. The yellow points with a red triangle inside are showing the dataset with one change point with a possible credible shift, that is supporting the event. The black points represent the control set used in the study conducted by Ön et al. (2021) and additionally the dataset obtained from the Zagros mountains by Andrews et al. (2020) (Table S3). Notably, these data points do not exhibit any indications of the purported climatic event. The purple dashed line represents the hypothetical boundary proposed by Ön et al. (2021), and it can be observed that the majority of the red triangles are located south of this line. Using triangles to represent the data that support the event on the map is aimed at enhancing their differentiation from the other data.

At first glance at Fig. 6, with some exceptions, the blue and yellow points (representing locations that show no credible change between 4.4 and 3.9 ka BP) cluster in and around Anatolia. In contrast, it is noteworthy that the red, and yellow points with triangles, which symbolize locations demonstrating a credible event with two or one supporting change points between 4.4 and 3.9 ka BP, tend to cluster in Italy, Greece, and the southern region below 35°N in Southwest Asia. Moreover, when the data locations used as a control set in Ön et al. (2021), which were asserted to be unaffected by the potential 4.2 ka BP climatic event without any statistical analysis, are included in Fig. 6 as black points, they exhibit a tendency to cluster with the blue points and around the hypothetical boundary with the yellow points. It is important to note that the Bayesian change point approach used in this study can only identify structural changes in individual time series and does not provide any information about geographical patterns. However, the hypothetical boundary proposed by Ön et al. (2021) remains relevant for this study (Fig. 6), despite potential differences in the results due to fundamental structural differences in the statistical approaches used in each study. Although this study uses different datasets and does not provide a temporal pattern as a conclusion, the contrasting regions identified by Di Rita et al. (2022) may be similar to those proposed by Ön et al. (2021) and this study. This suggests that the proposed spatial patterns may be consistent across studies, despite differences in the details and conclusions resulting from different approaches.

As noted by both Sirocko (2015) and Bini et al. (2019), the proposed asymmetrical bending of the ITCZ may be attributed to a major shift in multiple teleconnection patterns resulting from nonlinear interactions between the external and internal states of the climate system. As already highlighted by previous research (Finnè et al., 2011; Bini et al., 2019; Arz et al., 2015; Ön et al., 2021; Sirocko, 2015), it is crucial to emphasize the need for more data to achieve a comprehensive understanding of the distribution and extent of the event, not only in the studied region but also globally. This will aid in developing a better comprehension of the causes and mechanisms behind it.

4. Conclusions

The 4.2 ka BP event has been linked to various societal changes, including the collapse of several ancient civilizations, such as the Akkadian Empire in Mesopotamia and the Old Kingdom in Egypt. However, the Bayesian change point analysis results of this study align with those of Ön et al. (2021) and suggest that the impact of the event in southeast Europe and southwest Asia may have been more localized than previously thought, with a potential drought/cooling pattern observed in the southwestern portion of the studied region. An understanding of the spatial and temporal distributions of past climate events can aid in better comprehending their potential effects on ancient societies and civilizations.

The widespread acceptance of the 4.2 ka BP event as a global phenomenon has been called into question in recent literature, leading to a re-evaluation of the event's true geographical and temporal extent. This study, utilizing a test based on a Bayesian approach, highlights the importance of critical evaluation of proxy data and the need for a more nuanced understanding of past abrupt climate events. Additionally, the results provide new insights into the validity of the 4.2 ka BP event in southeast Europe and southwest Asia, and underscore the need for more high-resolution data from the region. Finally, I suggest the use of a Bayesian change point algorithm to identify abrupt events in paleoclimate time series as a powerful tool for paleoclimate research.

Credit author statement

I hereby declare that all of the conceptualization, data collection, literature review, methodology, data analysis, interpretation, and writing of the draft, with credit given to relevant sources, has been solely conducted by Z. Bora Ön.

Code availability

All the data and codes used to generate the results can be found at https://github.com/zboraon/changepointfor_4_2_kaBP_event.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All the data and codes used to generate the results can be found at https://github.com/zboraon/changepointfor_4_2_kaBP_event.

Acknowledgements

I used Veusz 3.5.3 (<https://veusz.github.io/>), an open-source scientific plotting package, to create all figures in the main text, with the exception of the map. The figures included in the supplementary material were generated using R and Veusz. The language of my work was edited and polished by OpenAI's ChatGPT, and subsequently reviewed and edited by Dr. Alan M. Greaves and finally by the author of this study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.quascirev.2023.108163>.

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